

HW2: Finite Element Analysis

1. **Acknowledgement if relevant:** e.g., significant help/assistance from other people, use of reference materials and/or AI tools etc.

2. **(Polynomial space $P^2(I)$ (6 points))**

Let $I = [x_1, x_2]$, with $x_3 = \frac{x_1+x_2}{2}$. Let $P^k(I)$ be the set of all polynomials of degree up to k . Define $\lambda_j \in P^2(I)$, $j = 1, 2, 3$, satisfying $\lambda_j(x_i) = \delta_{ij}$, $i, j = 1, 2, 3$. Show for any $p(x) \in P^2(I)$, it has the following expansion,

$$p(x) = p(x_1)\lambda_1(x) + p(x_2)\lambda_2(x) + p(x_3)\lambda_3(x).$$

Requirement: symbolic tools can not be used.

Note: this implies $\{\lambda_j\}_{j=1}^3$ is a basis for $P^2(I)$. Such basis is called the nodal basis. One does not need to find the explicit formula of each λ_j to complete the proof.

3. **(Approximation result / interpolation error) (14 points)**

Let $0 = x_0 < x_1 < \dots < x_N < x_{N+1} = 1$ be a mesh for $[0,1]$, and let $I_j = [x_{j-1}, x_j]$, $h_j = x_j - x_{j-1}$ for $j = 1, 2, \dots, N+1$, and $h = \max_{1 \leq j \leq N+1} I_j$. Consider

$$V_h = \{v \in C([0,1]) : v|_{I_j} \in P^2(I_j), j = 1, \dots, N+1\},$$

and for any given $u \in C([0,1])$, we define its piecewise quadratic interpolation $u_I \in V_h$, satisfying

$$u_I(x_j) = u(x_j), \quad \forall j = 0, 1, \dots, N+1,$$

and

$$u_I\left(\frac{x_{j-1} + x_j}{2}\right) = u\left(\frac{x_{j-1} + x_j}{2}\right), \quad \forall j = 1, 2, \dots, N+1.$$

Show that there exists a constant C , such that

$$\|u - u_I\|_{L^2(0,1)} \leq Ch^3 \|u'''\|_{L^2(0,1)}, \quad \forall u \in C^3([0,1]).$$

The positive constant C is independent of h and u .

4. **(Application of Lax-Milgram Theorem) (16 points)**

Let $V = H_0^1(0,1)$. Consider the following variational problem: look for $u(x) \in V$ that satisfies

$$a(u, v) = L(v), \quad \forall v(x) \in V, \tag{1}$$

where

$$a(u, v) = \int_0^1 p(x)u'(x)v'(x) + r(x)u'(x)v(x) + q(x)u(x)v(x)dx, \quad L(v) = (f, v). \tag{2}$$

Here $r(x) = 2$, $p(x) \geq p_0 > 0$, $q(x) \geq 0$. In addition $p(x), q(x) \in L^\infty(0,1)$, and $f(x) \in L^2(0,1)$. Here p_0 is a constant.

- i.) **(12 points)** Based on the Lax-Milgram theorem, prove the existence and uniqueness of the solution u in V . You are **required** to carry out your analysis using the H^1 norm $\|\cdot\|_{H^1(0,1)}$ for the space V . *You do not need to verify $a(\cdot, \cdot)$ is bilinear and $F(\cdot)$ is linear, but you do need to verify all the other conditions required in Lax-Milgram theorem in order to use the theorem.*

- ii.) (4 points) Note that as long as $r(x)$ is a constant, regardless it is 2 or not, your analysis in the previous i) will hold. Propose more general condition(s) on $r(x)$ so that the Lax-Milgram theorem can still be applied. For your proposed conditions, verify the coercivity and boundedness of $a(\cdot, \cdot)$.

5. **A surprise! (8 points)**

Consider the model problem

$$\begin{cases} -u''(x) = f(x), & x \in (0, 1), \\ u(0) = 1, \quad u(1) = 0, \end{cases} \quad (3)$$

its weak formulation: look for $u \in W$, such that

$$(u', v') = (f, v), \quad \forall v \in W, \quad (4)$$

and its finite element method: look for $U \in W_h$, such that

$$(U', v') = (f, v), \quad \forall v \in W_h. \quad (5)$$

Here

$$W = \{w \in C^0[0, 1] : w' \text{ is piecewise continuous and bounded, } w(0) = w(1) = 0\},$$

$$W_h = \{w \in W : w|_{I_j} \in P^1(I_j), j = 1, 2, \dots, N+1\}.$$

The same notation as in class is used for the finite element mesh. Let $G_i \in W$ satisfy

$$(v', G'_i) = v(x_i), \quad \forall v \in W, \quad (6)$$

where x_i is any given mesh node, $i = 1, 2, \dots, N$. Prove or verify that G_i is given by

$$G_i(x) = \begin{cases} (1 - x_i)x & 0 \leq x \leq x_i, \\ x_i(1 - x) & x_i \leq x \leq 1. \end{cases} \quad (7)$$

One also notes that $G_i \in W_h$. Finally by choosing $v = e = u - U$ in (6), show that

$$e(x_i) = (e', G'_i) = 0, \quad i = 1, \dots, N. \quad (8)$$

This implies U equals to u at mesh nodes. That is, U is the piecewise linear interpolant of u with respect to the mesh.

Additional notes: (i.) This somewhat surprising finding is a one-dimensional effect. Feel free to numerically verify it. (ii.) Based on the finding, one can establish the error estimate in max-norm: $\|u - U\|_{\max}$. (iii.) The function G_i is the Green's function for (3) associated with the delta function $\delta_{x_i}(x)$ at the node x_i .

6. **(Convection-diffusion equation, singularly perturbed problem) (26 points in total)**

Consider the following two-point boundary value problem

$$\begin{cases} -\varepsilon u''(x) + u'(x) = 0, & x \in (0, 1), \\ u(0) = 1, \quad u(1) = 0. \end{cases} \quad (9)$$

with a constant parameter $\varepsilon \in (0, 1]$. This is a convection-diffusion equation. When $\varepsilon \ll 1$, it is convection-dominated, and is a singularly perturbed problem.

- (i.) (**3 points**) Derive the exact solution $u(x)$.
- (ii.) (**6 points**) With the same notation as used in class (or in Question #2 above) for a mesh of $[0, 1]$ (i.e. x_j, I_j, h_j, N), formulate a C^0 - P^1 finite element method for this problem. Convert the method into an algebraic system. Particularly, document the algebraic form associated with the convective term $u'(x)$, which will be used in implementation.
- (iii.) (**12 points**) Implement the method (e.g. by adapting your code for HW1) for $\varepsilon = 0.5, 0.01, 0.001$ and when the mesh is uniform, that is $h_j = h = \frac{1}{N+1}$, with $N + 1 = 10, 10^2, 10^3, 10^4$.
- Plot numerical solutions against the exact solution, when $\varepsilon = 0.5$ and $N + 1 = 10$, and when $\varepsilon = 0.01$, with $N + 1 = 10, 10^2$, in 3 separate plots. *Just as in HW1, you would want to sample several points in each mesh element when plotting.*
 - Associated with each ε , tabulate the errors and orders in the L^2 norm (i.e. $\|\cdot\|_{L^2}$) and in the H^1 semi-norm (which is the same as the energy norm in HW1). The convergence order in the L^2 norm can be computed as

$$\frac{\log \left(\|e_h(\cdot)\|_{L^2} / \|e_{\frac{h}{10}}(\cdot)\|_{L^2} \right)}{\log(10)}.$$

Similarly, you can compute the convergence order in the H^1 semi-norm.

- (iv.) (**5 points + possible bonus**) Summarize your observation and comment on your results. Use your knowledge in mathematics and in numerical methods (including what you have learned so far about finite element methods) to provide some explanations. Feel free to run your codes with more values of ε and N for additional insight. As always for this course, computer codes are expected, to be attached to the very end of your entire report.

Here are some relevant references:

- Chapter 10: The Finite Element Method: Theory, Implementation, and Applications, Mats G. Larson, Fredrik Bengzon, 2013, Springer
- Chapter 9: Numerical Solution of Partial Differential Equations by the Finite Element Method, Claes Johnson, 1987, 2009, Dover